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Original article

The long-term clinical benefit and effectiveness of switching to once-daily quetiapine extended release in patients with schizophrenia

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Abstract

Objective:
To evaluate the long-term clinical benefit and effectiveness of switching to once-daily quetiapine extended release (XR) from an oral antipsychotic in patients with schizophrenia. Reasons for switching included insufficient efficacy, tolerability, and/or non-acceptability. The primary endpoint was the percentage of patients achieving an improvement in Clinical Global Impression – Clinical Benefit (CGI-CB) scale scores.

Research design and methods:
A 24-week, international, multicentre, open-label, prospective study (www.clinicaltrials.gov: NCT00640601). After a 7–14 day enrolment period (depending whether prior antipsychotic mono- or combination therapy), all patients received quetiapine XR 300 mg once daily (day 1), 600 mg/day (day 2), 600–800 mg/day (day 3) and 400–800 mg/day thereafter, with down-titration and discontinuation of prior antipsychotic by day 4.

Results:
A total of 62% of patients completed the study and 56.9% (LOCF, ITT) achieved a significant improvement in CGI-CB (95% CI [0.51, 0.63]; $p = 0.02$). Switches due to insufficient efficacy showed a significant improvement (60%, 95% CI [0.51, 0.68]; $p = 0.02$), compared to 54.4% ([0.44, 0.64]; $p = 0.38$) and 52.4% ([0.36, 0.68]; $p = 0.76$) of switches due to insufficient tolerability and non-acceptability respectively (both $p =$ ns). Patients previously on olanzapine and quetiapine IR showed a significant improvement in CGI-CB (62.6% [$p = 0.02$] and 61.2% [$p = 0.04$], respectively). Somnolence (18.0%) and dizziness (14.6%) were the main adverse events. Anticholinergic use decreased from 7.1 to 2.7%. Overall mean weight gain was 0.4 kg; 12.9% of patients experienced a weight gain of $\geq$7% and 15% experienced a clinically relevant shift in triglycerides from baseline.

Conclusions:
A majority of patients switched from other antipsychotics to quetiapine XR experienced clinical benefit. This was supported by all other efficacy outcomes regardless of the reason for switching. Safety data confirmed quetiapine XR was safe and well tolerated. The open-label design and lack of a placebo group represent limitations.

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Introduction

Antipsychotics are recognized as an essential component of treatment for most patients with schizophrenia[1–4]. Despite offering a similar level of efficacy to conventional or first generation antipsychotics (FGA)[5], response to atypical or second generation antipsychotics (SGA) is variable in terms of both efficacy and tolerability[6]. Thus, it is common in clinical practice for patients to switch from one antipsychotic to another, both within and across different classes, as was first demonstrated in phase 1 of the Clinical Antipsychotic Trials of Intervention Effectiveness (CATIE) study in which 74% of patients switched from their initial antipsychotic in the first 18 months of treatment[7]. In this study, designed to reflect real-world practice, the primary outcome for the trial was discontinuation of antipsychotic treatment for any reason, including lack of efficacy, side-effects and patient choice. Other studies of similar design in first episode patients have also reported comparably high rates of switching[8,9]. The decision to change a treatment regimen is also influenced by factors related to drug formulation, dosing and titration – all of which may affect patients' satisfaction and acceptability of treatments, adherence and ultimately outcome[10,11].

Quetiapine fumarate extended release (quetiapine XR) is a formulation that allows for the convenience of a once-daily dosing regimen with a simpler dose titration schedule, which may increase patients' adherence to treatment[12]. Together with its demonstrated safety profile, the greater efficacy afforded by treatment with quetiapine XR compared to placebo on positive and negative symptoms in patients with schizophrenia has previously been demonstrated in most[12–15] but not all randomized clinical trials[16]. Notably, patients with stable schizophrenia were successfully switched from twice daily quetiapine immediate release (quetiapine IR) to the once-daily XR formulation without compromising safety or tolerability[17]. In addition, a 12 week study showed that more than half of patients with schizophrenia who had inadequate response to prior antipsychotic treatment demonstrated improvements on the Clinical Global Impression – Clinical Benefit (CGI-CB) scale after switching to once-daily quetiapine XR (62.8%; 95% confidence interval [CI]: 58.4–67.1; $p<0.0001$)[18]. Switching to quetiapine XR has also been associated with improved quality of life, prevention of relapse, and symptomatic remission in patients with schizophrenia[15,19,20]. Taken together, this body of data suggests that quetiapine XR may be an effective and safe agent for the treatment of schizophrenia[21], but its long-term efficacy for patients with inadequate response to their current medication remains to be established.

The primary objective of this 24 week study (D1443L00025) was to demonstrate the clinical benefit of once-daily quetiapine XR after switching from prior oral antipsychotic treatment in patients with schizophrenia. More specifically, the present study expanded the time frame of evaluation and to a naturalistic, clinically relevant setting. Secondary objectives included the evaluation of quetiapine XR for other efficacy outcomes, safety, tolerability, remission and patient-reported assessments of overall functioning and treatment satisfaction.

Patients and methods

Patients

Female and male outpatients aged ≥18 and ≤65 years who fulfilled criteria for schizophrenia (any subtype) according to the Diagnostic and Statistical Manual of Mental Disorders, 4th edition, Text Revision (DSM-IV TR) were eligible for

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enrolment and recruited from a specialist care setting. All enrolled patients had experienced inadequate treatment with their prior antipsychotic medication, due to insufficient efficacy (residual positive or negative symptoms, aggression/hostility, or cognitive impairment), insufficient tolerability (adverse effects), and/or non-acceptability (dissatisfaction with treatment including dosage regimen). Non-acceptability was based on the subject's and/or principal investigator's decision and for reasons other than adverse effects. Exclusion criteria included: first schizophrenic episode, antipsychotic-naïve patients; combination therapy less than 7 days before initiation of study medication; intolerance to quetiapine IR or on quetiapine once-daily; known lack of response to clozapine or treatment with clozapine within 4 weeks of enrolment (previous history of clozapine treatment was not exclusionary); treatment with a long-acting antipsychotic (LAI) within one dosing interval; evidence of clinically relevant disease or abnormal laboratory test results; and treatment with potent inhibitors or inducers of cytochrome P450 3A4 enzymes.

Ethics approval for the study was obtained from an independent, central, ethics review board (Optimum Clinical Research Inc. Ethics Review Board, Oshawa, Ontario, Canada), local hospital ethics boards, university research ethics boards, and provincial Canadian Colleges of Physicians and Surgeons review committees where appropriate. Full, informed, written consent was obtained from each subject at study entry. The study was performed in accordance with current International Conference on Harmonization Good Clinical Practice (ICH/GCP) guidelines and the Declaration of Helsinki and its subsequent revisions.

The Clinical Trial Application (CTA) for the study was approved by Health Canada on January 16, 2008 and the first patient was enrolled on March 14, 2008; the last patient completed the study on July 8, 2010.

Study design

This study (www.clinicaltrial.gov: NCT00640601) was a 24 week multicentre, open-label study conducted in Canada, Australia, Hong Kong and Republic of Korea within several months of commercialization of quetiapine XR (with the exception of Republic of Korea where the study was initiated prior to launch). Treatment with quetiapine XR was preceded by a 7 day enrolment period (or a 14 day enrolment period for patients on combination therapy to taper down to one antipsychotic), followed by a cross-titration period of 4 days, and a 24 week (168 days) flexible dosing period. Quetiapine XR (200 or 300 mg tablets) was administered orally once daily, preferably in the evening. During the cross-titration period (days 1–4), patients started on 300 mg/day, and were titrated up to 600 mg/day by day 2, and 600–800 mg/day by day 3, depending on clinical judgment. On day 4, the 800 mg dose could be maintained or, if not tolerated, the dose could be decreased to a target dose of 600 or 400 mg/day. Simultaneously, patients were down-titrated from their original antipsychotic treatment, to reach a target dose of 0 mg by day 4. Patients on quetiapine IR were allowed to be switched directly to the equivalent dose of quetiapine XR. For the remaining 24 week treatment period, a flexible once-daily dose of quetiapine XR between 400 mg and 800 mg was permitted with minimum dose adjustments of 200 mg.

Concomitant medication

Concomitant medications such as anxiolytics, antidepressants and mood stabilizers were allowed if taken for at least 2 weeks at stable dose before starting the study. No changes in concomitant medications were permitted during the study with the exception of approved medications (to specified maximum doses) for sleep if required. These included zolpidem tartrate (maximum dose = 10 mg/day), chloral hydrate (maximum dose = 2 g/day), zaleplon (maximum dose = 20 mg/day) and zopiclone (maximum dose = 7.5 mg/day).

Outcome measures

Primary endpoint

The primary endpoint was the proportion of patients who achieved an improvement (decrease) in the CGI-CB scale by week 24. The CGI-CB scale or CGI–Efficacy Index, is the third global subscale developed from the CGI[22], that assesses a composite of clinically relevant changes from baseline in terms of efficacy and tolerability due to treatment (Table 1). The scale ranges from a score of 1 (greatest benefit/least burden from treatment), to 10 (least benefit/greatest burden from treatment). The primary hypothesis was that at least 50% of patients would experience an improvement (decrease) in CGI-CB scores by study end. The CGI and derived subscales including the CGI-CB have been used as primary outcome measures in medication studies[18]. The CGI has been shown to be sensitive in the hands of experienced clinicians particularly for relative change and correlates well with the Positive and Negative Syndrome Scale (PANSS)[23,24].

Secondary endpoints

Secondary endpoints included the change from baseline to week 24 for the following scales: CGI-CB; PANSS total score, and positive, negative and general psychopathology subscale scores; Clinical Global Impression–Improvement (CGI-I); Clinical Global Impression–Severity (CGI-S); Global Assessment Scale (GAS); and Social and

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Table 1. Assessment of clinical benefit using the CGI-CB scale (CGI–Efficacy Index)[22].

Burden of adverse events				
Therapeutic effect	None	No significant interference	Significant interference	Outweighs therapeutic effect
Marked	1	2	5	10
Moderate	3	4	6	10
Minimal	7	8	9	10
Unchanged/worse (no effect)	10	10	10	10

Rank 1 = most benefit from treatment.
Rank 10 = no benefit from treatment.

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Occupational Functioning Assessment Scale (SOFAS). Other exploratory analyses and measures of patients' overall functioning were used within this context to further assess the effects of quetiapine XR and will be reported in detail in a separate publication.

Additional endpoints

Analyses of AEs including serious AEs (SAEs) leading to drug discontinuation, adherence with the protocol, vital signs (blood pressure [BP] and heart rate [HR]) and electrocardiogram (ECG) were included in safety data. The tolerability of the study medication was also assessed with the use of the Barnes Akathisia Rating Scale (BARS) and Simpson–Angus Scale (SAS) for severity of extrapyramidal symptoms (EPS). Finally, haematology, clinical chemistry, and metabolic (fasting plasma glucose [FPG], insulin, and lipids; glycosylated haemoglobin [$HbA1_C$]), weight, waist-to-hip ratio [WHR], body mass index [BMI], and proportion of patients with ≥7% weight gain) parameters were reported using descriptive statistics.

Statistical analysis

The intent-to-treat (ITT) population comprised the population for the primary analysis. Mean change from baseline for the primary analysis was assessed with at least one non missing post-baseline assessment. Missing values for the primary analysis were handled using the last observation carried forward (LOCF) approach. The confidence interval for the proportion of patients achieving an improvement in CGI-CB was computed using the asymptotic normal approximation to the binomial distribution. The corresponding one-sided *p*-value (based on the *a priori* hypothesis that patients would experience an improvement in CGI-CB scores) was also presented. For a consistency check, the primary analysis was repeated using the per-protocol (PP) population (LOCF and the completers). Subgroup analyses were also performed based on the reason for switching and prior antipsychotic medication. The sample size calculation was based on the ability to detect a difference of 50% in the primary outcome variable, CGI-CB, for the population of completers. Given a true proportion of at least 58%, a sample size of 295 treated patients was estimated to have 85% power to detect a change in the primary outcome, at a one-sided significance level of 0.025. To account for withdrawals, a total of 331 patients were recruited. Data were reported as means (standard deviation, SD).

For the secondary endpoints, an analysis of covariance (ANCOVA) model with mixed effects was used for the change from baseline to the end of treatment. Fixed effects included baseline values. Analyses of the proportion of patients with a CGI-S score of ≤3 at week 24 and the proportion of patients using anticholinergic medication after administration of the first dose of quetiapine XR during the study were also performed.

Results

Patients

A total of 300 patients were enrolled in this trial; five withdrew from the study before the administration of quetiapine XR: three due to voluntary discontinuation, one due to an AE and one additional subject was lost to follow-up. Patients (who did receive quetiapine XR) were enrolled from Canada ($n = 227$; 77%), South Korea ($n = 35$; 12%), Hong Kong ($n = 27$; 9%) and Australia ($n = 6$; 2%). Among this study population, 138 (46.8%) patients were enrolled due to insufficient efficacy, 112 (38.0%) due to insufficient tolerability and 44 (14.9%) due to non-acceptability, while one subject (0.3%) did not report any reason. The patients were also subdivided based on prior medications: 91 (31%) patients had been on olanzapine (mean daily dose [SD] = 12.53 mg [6.58]); 86 (29%) on quetiapine IR (mean daily dose [SD] = 544 mg [219.6]); 64 (22%) on risperidone (mean daily dose [SD] = 3.1 mg [1.6]); 28 (9%) on other SGAs, 24 (8%) on FGAs, and 2 (1%) who were on combination therapy, antipsychotic-naïve or who had discontinued before obtaining mono-therapy data. The 295 patients who received at least one dose of the study medication were included in the ITT population, and 183 (62%) patients completed the study (Figure 1).

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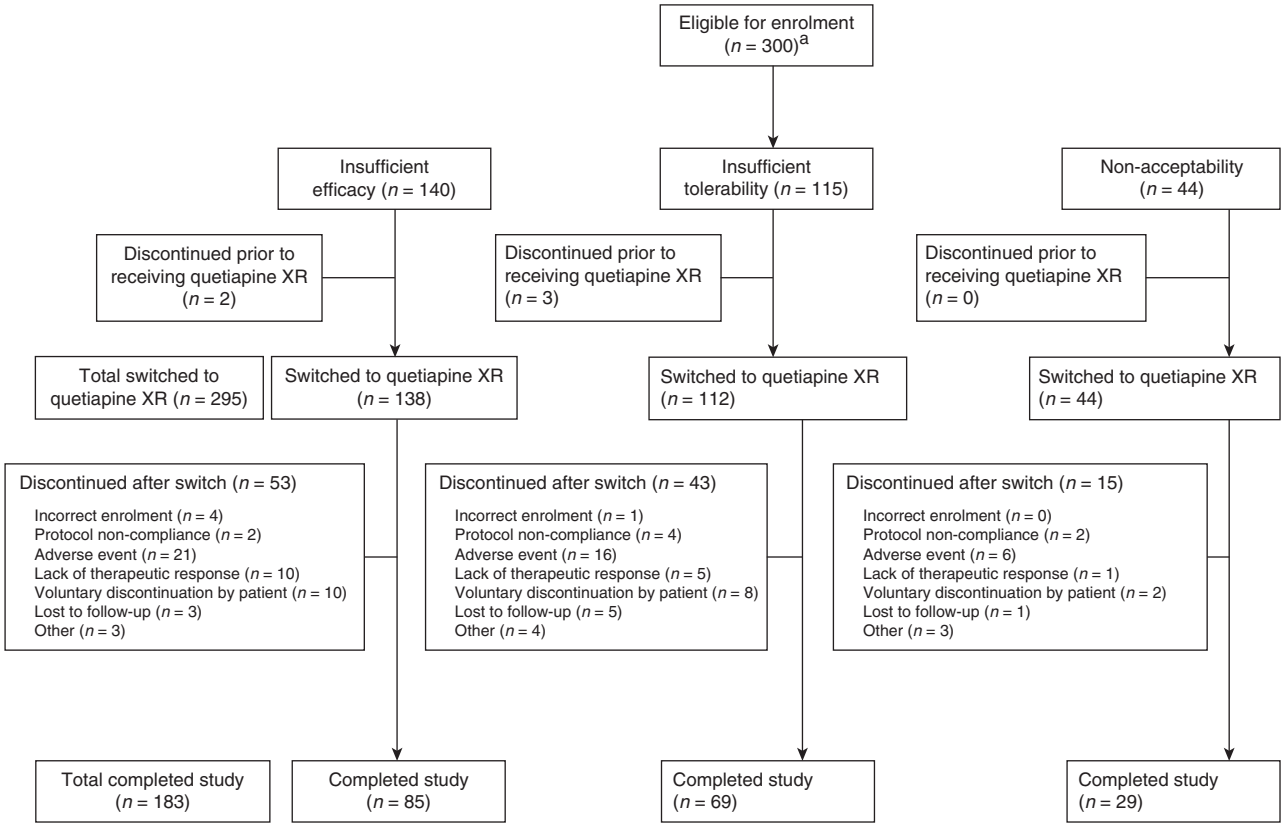

Figure 1. Patient disposition during the study.
[a]One patient with no reported reason for switching to quetiapine XR is included in the total switched to quetiapine XR.

Baseline demographics and clinical characteristics were balanced between subgroups whether patients were divided by reasons for switching to quetiapine XR (Table 2) or by prior antipsychotic treatment (data not shown). Amongst the 138 patients who switched to quetiapine XR due to insufficient efficacy, the main reason cited for switching (could be reported more than once) was persistent positive symptoms (72.5%), followed by negative symptoms (59.4%) and general psychopathology (50.0%). Amongst the 112 patients who switched to quetiapine XR due to insufficient tolerability, 41.1% switched due to weight gain, followed by other reasons (31.3%), sexual dysfunction (8.0%) and anticholinergic side effects (4.5%).

Treatment

During the study, the average dose (SD) of quetiapine XR was 558.6 mg (311.2) daily, which was similar between subgroups. Average doses ranged from 545.5 mg (294.3) to 571.9 mg (357.0) daily in subgroups based on reason for switching, and from 470.8 mg (146.3) to 615.4 (773.8) based on prior medication.

Throughout the study, the proportion of patients considered fully adherent (defined as taking study drug as directed 80–100% of the time) was 80.0%. Overall adherence was numerically lower in patients who had switched due to non-acceptability of their prior medication (65.9% were considered fully adherent), while adherence was high and similar in the insufficient efficacy and tolerability subgroups (82.6% and 82.1%, respectively).

Efficacy

Overall, a majority of patients ($n = 157/276$, 56.9%) demonstrated a significant improvement in their CGI-CB score (95% CI [0.51, 0.63]; $p = 0.02$). Patients who switched to quetiapine XR due to insufficient efficacy showed a significant improvement (60%, 95% CI [0.51, 0.68]; $p = 0.02$), compared to 54.4% ([0.44, 0.64]; $p = 0.38$) and 52.4% ([0.36, 0.68]; $p = 0.76$) of patients who switched due to insufficient tolerability and non-acceptability, respectively (both $p =$ ns) (Figure 2). Patients who completed the full 24 week study period showed a more pronounced improvement in CGI-CB score, with 70.5% of completers experiencing a significant improvement ([0.63, 0.77], $p < 0.0001$). When patients were evaluated based on their prior antipsychotic medication, the proportion with CGI-CB improvement ranged from 34.8% to 62.6% for those who switched from FGAs

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Table 2. Patient demographic and clinical characteristics at baseline by reason for switching to quetiapine XR.

	Overall ($n=295$)[a]	Insufficient efficacy ($n=138$)	Insufficient tolerability ($n=112$)	Non-acceptability ($n=44$)
Gender, *n* (%)				
Male	182 (61.7)	93 (67.4)	60 (53.6)	28 (63.6)
Female	113 (38.3)	45 (32.6)	52 (46.4)	16 (36.4)
Age, years (SD)	37.8 (12.1)	38.4 (12.0)	36.5 (12.0)	38.8 (12.9)
Race, *n* (%)				
White	185 (62.7)	88 (63.8)	68 (60.7)	28 (63.6)
Black	8 (2.7)	5 (3.6)	2 (1.8)	1 (2.3)
Asian	86 (29.2)	37 (26.8)	36 (32.1)	13 (29.5)
Aboriginal	6 (2.0)	3 (2.2)	2 (1.8)	1 (2.3)
Other	10 (3.4)	5 (3.6)	4 (3.6)	1 (2.3)
Marital status, *n* (%)				
Single	224 (76.2)	104 (75.9)	88 (78.6)	31 (70.5)
Married	30 (10.2)	16 (11.7)	8 (7.1)	6 (13.6)
Common law	12 (4.1)	3 (2.2)	6 (5.4)	3 (6.8)
Divorced	27 (9.2)	13 (9.5)	10 (8.9)	4 (9.1)
Widowed	1 (0.3)	1 (0.7)	0 (0.0)	0 (0.0)
Employment status, *n* (%)				
Full-time employed	28 (9.5)	10 (7.2)	11 (9.8)	7 (15.9)
Part-time employed	43 (14.6)	16 (11.6)	19 (17.0)	8 (18.2)
House-person/student	25 (8.5)	11 (8.0)	11 (9.8)	3 (6.8)
Retired/long term sick-leave	37 (12.5)	16 (11.6)	12 (10.7)	9 (20.5)
Unemployed	162 (54.9)	85 (61.6)	59 (52.7)	17 (38.6)
Psychiatric history				
Disease duration, years (SD)	11.0 (10.4)	11.6 (10.3)	10.5 (10.8)	9.3 (8.7)
Number of schizophrenic episodes over lifetime, *n* (SD)	4.5 (6.4)	4.4 (4.5)	4.1 (4.7)	6.0 (12.6)
Presence of family members with schizophrenia, *n* (%)	83 (28.5)	37 (27.4)	36 (32.4)	10 (22.7)
DSM-IV diagnosis, *n* (%)				
Disorganized	7 (2.4)	5 (3.6)	1 (0.9)	1 (2.3)
Catatonic	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
Paranoid	248 (84.1)	111 (80.4)	98 (87.5)	39 (88.6)
Undifferentiated	40 (13.6)	22 (15.9)	13 (11.6)	4 (9.1)
Mean (SD) baseline scores				
CGI-CB score	6.3 (2.6)	6.8 (2.4)	6.1 (2.7)	5.4 (2.6)
CGI-I score	3.5 (0.8)	3.5 (0.8)	3.6 (0.8)	3.5 (0.8)
CGI-S score	3.8 (1.2)	4.3 (1.0)	3.3 (1.2)	3.5 (1.2)
PANSS total score	74.5 (22.3)	82.1 (22.1)	65.1 (19.3)	73.7 (23.1)
GAS score	53.8 (14.9)	49.1 (14.4)	59.4 (14.3)	55.5 (14.9)
SOFAS score	54.1 (14.8)	49.7 (14.9)	59.1 (14.3)	55.5 (11.2)

[a]One patient with no reported reason for switching to quetiapine XR is included in the Overall column.

and olanzapine, respectively. Notably, only patients previously on olanzapine and quetiapine IR showed a significant improvement on CGI-CB (62.6% [$p=0.02$] and 61.2% [$p=0.04$], respectively). Consistent with the analysis of the ITT population the results of PP analysis also demonstrated significant improvement for primary efficacy measures (LOCF and completers).

Secondary efficacy outcomes

Improvements in CGI-CB, CGI-I, and CGI-S scores were significant in the overall study population ($p<0.0001$) at week 24 (Table 3). Importantly, improvements in CGI-CB scores occurred as early as week one for 27.5% of patients, and 56.9% of patients at week 24. Significant improvements in CGI-CB scores were observed when patients were assessed based on reasons for switching to quetiapine XR ($p<0.05$ for all subgroups), but only those patients who switched due to insufficient efficacy or tolerability also showed a significant improvement on CGI-S and CGI-I scores (both $p<0.05$). In subgroup analyses based on prior medication, CGI-CB scores decreased significantly for patients switched from olanzapine (−1.7), quetiapine IR (1.45) and from other SGAs (−2.52) (all $p<0.05$). Furthermore, CGI-S scores decreased significantly for patients switched from olanzapine (−0.71), risperidone (−0.59) and quetiapine IR (−0.31) (all $p<0.01$).

Total PANSS score significantly improved in the overall population, regardless of reason for switching to quetiapine XR or prior antipsychotic medication ($p<0.001$). The overall population also showed significant decreases in PANSS positive symptoms, negative symptoms and general psychopathology subscales ($p<0.001$; Figure 3). On all PANSS subscale scores, improvements tended to be numerically larger in the insufficient efficacy subgroup compared to the other subgroups. Nonetheless, patients who switched due to insufficient tolerability still showed

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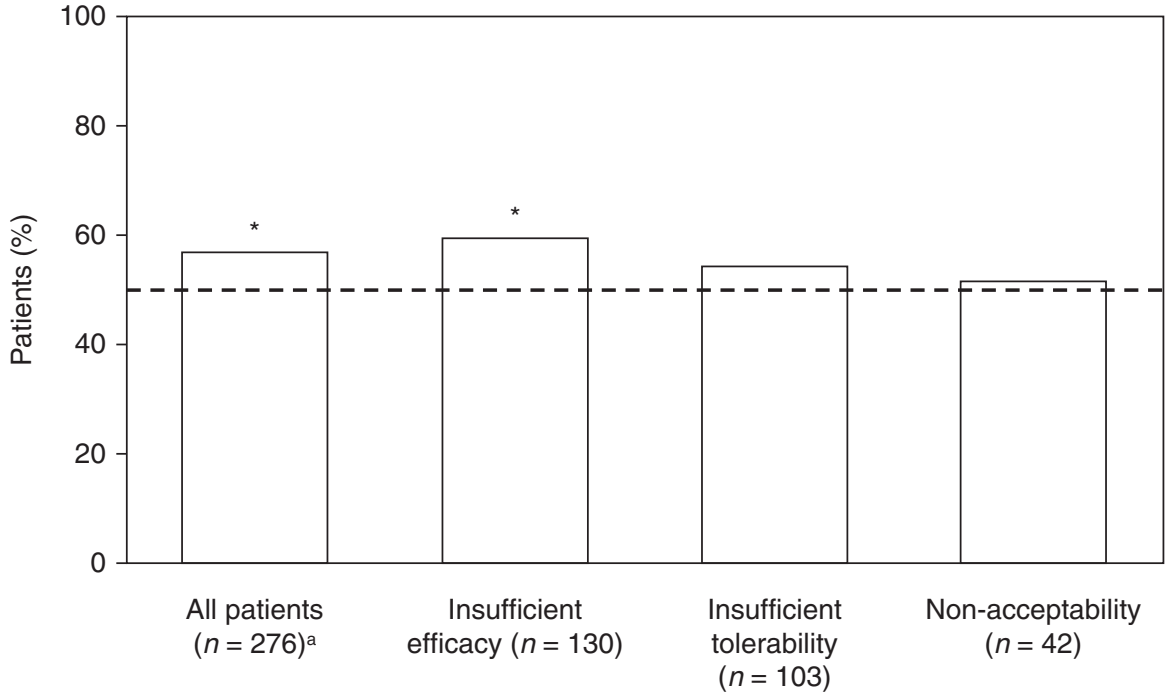

Figure 2. Improvement in CGI-CB from baseline to week 24 after switching to quetiapine XR.
*$p<0.05$, one-sided p-value from a binomial test, with a null hypothesis that ≤50% patients would achieve an improvement in CGI-CB (numbers for each subgroup are based on those with non-missing information). [a]One patient with no reported reason for switching to quetiapine XR is included in the total switched to quetiapine XR.

Table 3. Change in CGI scores from baseline to week 24 (LOCF) by reason for switching to quetiapine XR.

	Overall[a]	Insufficient efficacy	Insufficient tolerability	Non-acceptability
CGI-CB				
n	276	130	103	42
Mean (SD)	−1.41 (3.84)***	−1.66 (3.79)***	−1.20 (4.13)**	−1.07 (3.29)*
CGI-I				
n	254	119	94	40
Mean (SD)	−0.55 (1.56)***	−0.66 (1.51)***	−0.53 (1.70)**	−0.25 (1.34)
CGI-S				
n	276	130	103	42
Mean (SD)	−0.51 (1.17)***	−0.81 (1.15)***	−0.27 (1.12)*	−0.21 (1.20)

[a]One patient with no reported reason for switching to quetiapine XR is included in the Overall column.
*$p<0.05$; **$p<0.01$; ***$p<0.0001$.

significant improvements in PANSS negative symptoms and general psychopathology subscales (all $p<0.001$), but not on the positive symptoms subscale.

Symptomatic remission was assessed using the Andreasen criteria (defined as a score of ≤3 on the PANSS scale items P1, P2, P3, N1, N4, N6, G5, and G9)[25]. At baseline, 22% of patients met criteria for symptomatic remission and this proportion increased to 43% at end of the study (ITT). These results will be reported in detail in a separate publication. GAS scores increased by an average of 5.54 (4.87) points in the overall population ($p<0.0001$) by week 24. This improvement ranged from 1.67 to 8.55 points in subgroups based on reason for switching ($p<0.05$ for insufficient efficacy and intolerability subgroups and all prior medication subgroups, except for patients who had been treated with FGAs). Importantly, a large majority of patients (84%) had stable or improved GAS scores over the course of the study.

Similarly, SOFAS scores increased by an average of 6.45 (13.82) points in the overall population ($p<0.0001$), and improved or were stable in 87% of patients over the course of the study. When patients were assessed by reason for switching to quetiapine XR, all subgroups showed a significant improvement ranging from 1.98 to 9.36 points ($p<0.0001$ for insufficient efficacy and tolerability subgroups). Improvements on the

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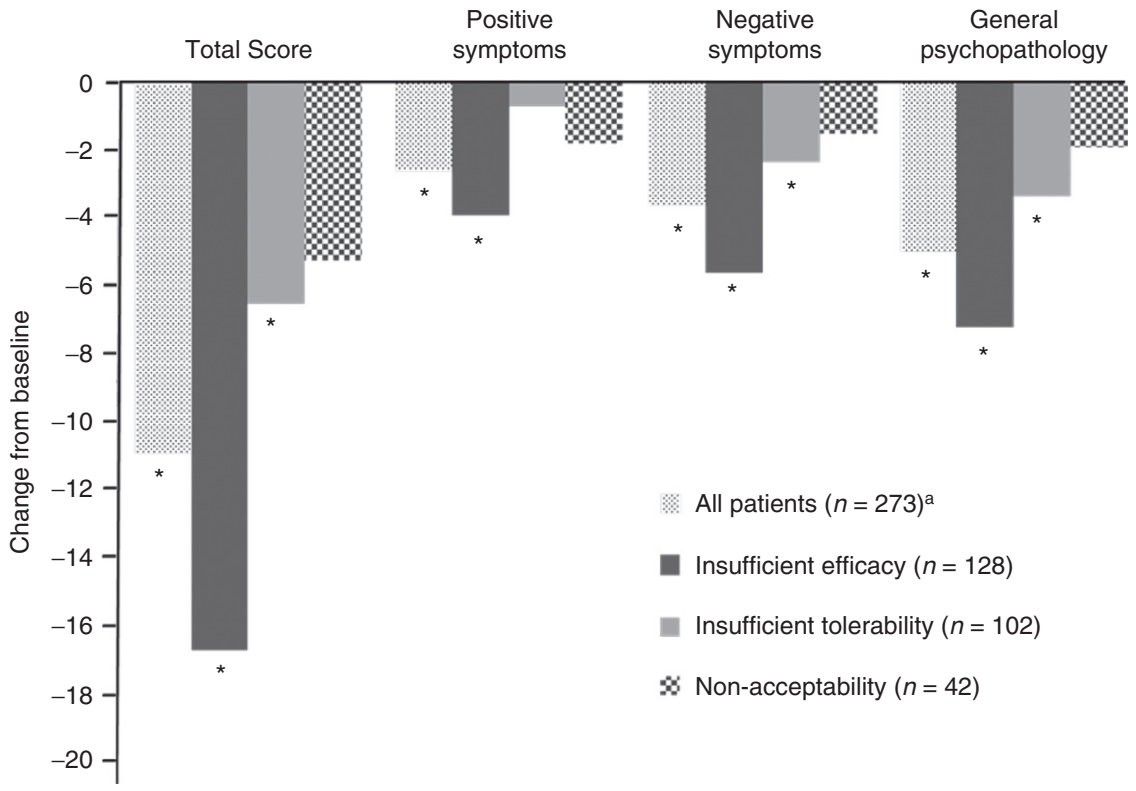

Figure 3. Change in PANSS total and subscale scores from baseline to week 24 after switching to quetiapine XR.
*$p<0.001$. [a]One patient with no reported reason for switching to quetiapine XR is included in the total switched to quetiapine XR.

SOFAS scale were significant regardless of prior antipsychotic medication ($p<0.05$).

Tolerability

AEs

Overall, quetiapine XR was well tolerated by all patients. Approximately three quarters of patients ($n=224/295$, 75.9%) experienced at least one AE during the course of the study. Somnolence (18.0%) and dizziness (14.6%) were the main AEs reported (Table 4). Serious AEs were reported in 11.2 % ($n=33/295$) of patients with the majority (79%) being related to an exacerbation of psychosis. There were no deaths during the study. There were no differences between the subgroups with regards to SAEs and/or AEs. A total of 112 (38.0%) patients discontinued quetiapine XR during the study. The most common reasons for discontinuing the study medication are listed in Figure 1.

EPS

There was a low incidence of EPS with SAS and BARS scores either unchanged or improved in 89.1% and 91.8% of patients at study completion, respectively. Improvements were noticeable in all subgroups, although patients who switched due to non-acceptability were less likely to show improvements on either the SAS or BARS (Table 5).

A total of 7.1% of patients were using anticholinergic medication on day 1 and this proportion decreased to 2.7% by the end of the study (including patients who withdrew early). Patients in the non-acceptability (13.6%) and insufficient tolerability (11.6%) subgroups had the highest rates of anticholinergic use at baseline, decreasing to 6.9% and 2.9% respectively by week 24.

Laboratory data, vital signs and physical findings

No clinically significant differences between baseline and week 24 assessments were observed with respect to haematology, BP, HR, and ECG parameters. Among those patients who had normal values at baseline, 3.6%, 1.0%, 6.4%, 5.8%, 13.4%, and 15.0% showed clinically significant shifts after the study period for FPG, HbA1c, total cholesterol, low-density cholesterol, high-density cholesterol, and triglycerides, respectively (Table 6).

The mean BMI at baseline was 28.7 (6.89) kg/m^2 and 67.8% of patients were overweight, obese or severely obese by BMI. Although 12.9% of patients experienced a weight gain of ≥7%, the overall mean weight gain was only 0.4 kg (5.16) with a mean change in WHR of 0.01 (0.07). There were no differences between subgroups for patients with a weight gain of ≥7% from baseline. Further, switching to

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Table 4. Most frequent (≥5%) adverse events while on study medication by preferred term, system organ class and reason for switching to quetiapine XR.

Preferred term[a]	Overall ($n=295$)[b]		Insufficient efficacy ($n=138$)		Insufficient tolerability ($n=112$)		Non-acceptability ($n=44$)	
n (%)								
Somnolence	53	(18.0)	21	(15.2)	28	(25.0)	4	(9.1)
Dizziness	43	(14.6)	23	(16.7)	17	(15.2)	3	(6.8)
Sedation	42	(14.2)	25	(18.1)	14	(12.5)	3	(6.8)
Dry mouth	38	(12.9)	20	(14.5)	17	(15.2)	1	(2.3)
Anxiety	27	(9.2)	13	(9.4)	9	(8.0)	5	(11.4)
Constipation	27	(9.2)	12	(8.7)	12	(10.7)	3	(6.8)
Headache	26	(8.8)	6	(4.3)	15	(13.4)	5	(11.4)
Insomnia	24	(8.1)	12	(8.7)	9	(8.0)	3	(6.8)
Fatigue	22	(7.5)	15	(10.9)	6	(5.4)	1	(2.3)
Weight increased	22	(7.5)	11	(8.0)	10	(8.9)	1	(2.3)
Nausea	17	(5.8)	7	(5.1)	8	(7.1)	2	(4.5)

[a]Patients with multiple events in the same category are counted only once in that category.
[a]Patients with events in more than one category are counted once in each of those categories.
[b]One patient with no reported reason for switching to quetiapine XR is included in the Overall column.

Table 5. Categorical changes in SAS and BARS scores from baseline to week 24/early withdrawal (LOCF) by reason for switching to quetiapine XR.

	Overall[a]	Insufficient efficacy	Insufficient tolerability	Non-acceptability
SAS				
n	269	127	100	41
Improving: *n* (%)	124 (46.1)	63 (49.6)	45 (45.0)	15 (36.6)
No change: *n* (%)	117 (43.5)	47 (37.0)	50 (50.0)	20 (48.8)
Worsening: *n* (%)	28 (10.4)	17 (13.4)	5 (5.0)	6 (14.6)
BARS				
n	267	126	99	41
Improving: *n* (%)	75 (28.1)	44 (34.9)	27 (27.3)	4 (9.8)
No change: *n* (%)	170 (63.7)	67 (53.2)	69 (69.7)	33 (80.5)
Worsening: *n* (%)	22 (8.2)	15 (11.9)	3 (3.0)	4 (9.8)

[a]One patient with no reported reason for switching to quetiapine XR is included in the Overall column.

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quetiapine XR induced either no or minimal weight gain in all subgroups (−0.6 to 1.7 kg over 24 weeks) and even decreased weight in the subgroup of patients switching from olanzapine (−0.6 kg).

Discussion

The main goal of this international clinical trial was to investigate whether patients with a diagnosis of schizophrenia demonstrating inadequate response, intolerability, or non-acceptability to their current oral antipsychotic medication would experience clinical benefit from switching to quetiapine XR. The primary outcome, a significant improvement (decrease) in CGI-CB score in 50% or more of patients, was achieved ($p<0.05$). This result suggests that quetiapine XR may be an effective therapeutic option for patients with schizophrenia, and/or their caregivers or physicians, who are dissatisfied with their current treatment. While at least 50% of patients in all subgroups showed an improvement on the CGI-CB scale ($p<0.05$ for all subgroups), switching to quetiapine XR had the greatest benefit among those who had switched due to insufficient efficacy, with 60.0% showing an improvement on their CGI-CB score ($p<0.05$). The lack of statistical significance in the other subgroups (i.e., switched due to insufficient tolerability and non-acceptability) may be due, at least in part, to the smaller sample sizes of those subgroups. Patients who stayed on quetiapine XR for the full 24 week study period were more likely to present improvements on the CGI-CB scale (70.5%, $p<0.0001$). The current data are not only consistent with a previous study[18], but also expand the time period studied (24 weeks vs. 12 weeks), which suggests that quetiapine XR can potentially offer a longer term solution for patients with schizophrenia whose current medication is not satisfactory.

When subgroups were analyzed based on prior antipsychotic medication, a majority (>50%) of patients who were previously treated with olanzapine and quetiapine IR showed improvements on the CGI-CB score (both $p<0.05$). Once again, the lack of statistical significance in the other subgroups may be due to the smaller sample sizes. Notably, 64.0% of patients previously treated with other SGAs improved on their CGI-CB score, the

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Table 6. Clinical laboratory parameters and body weight at baseline and mean changes in parameters to treatment end (week 24 or early withdrawal) by reason for switching to quetiapine XR.

Parameters	Overall ($n=295$)[a,b,c]	Insufficient efficacy ($n=138$)[a,b]	Insufficient tolerability ($n=112$)[a,b]	Non-acceptability ($n=44$)[a,b]
Glucose (fasting)[d], mmol/L				
Baseline	5.46 (1.06)	5.60 (1.28)	5.26 (0.78)	5.55 (0.89)
Change	0.09 (0.78)	0.09 (0.71)	0.19 (0.90)	−0.15 (0.68)
Proportion of patients with clinically relevant shift (≥7 mmol/L)	3.6%	3.9%	2.8%	5.1%
HbA1C				
Baseline	5.48 (0.64)	5.50 (0.73)	5.41 (0.55)	5.55 (0.55)
Change	0.12 (0.32)	0.10 (0.26)	0.16 (0.39)	0.11 (0.35)
Proportion of patients with clinical relevant shift (>7.5%)	1.0%	0.7%	1.8%	0%
Total cholesterol, mmol/L				
Baseline	5.13 (1.17)	5.13 (1.26)	5.11 (1.04)	5.19 (1.20)
Change	0.03 (0.78)	0.05 (0.87)	0.08 (0.71)	−0.16 (.62)
Proportion of patients with clinically relevant shift (≥6.21 mmol/L)	6.4%	5.6%	8.5%	2.9%
Low-density cholesterol, mmol/L				
Baseline	3.13 (1.04)	3.12 (1.08)	3.09 (0.99)	3.28 (1.03)
Change	−0.05 (0.71)	−0.05 (0.76)	0.03 (0.69)	−0.22 (0.61)
Proportion of patients with clinically relevant shift, (≥4.2 mmol/L)	5.8%	6.2%	6.5%	2.9%
High-density cholesterol, mmol/L				
Baseline	1.20 (0.34)	1.18 (0.36)	1.23 (0.34)	1.19 (0.26)
Change	−0.01 (0.21)	−0.01 (0.21)	−0.02 (0.23)	0.00 (0.15)
Proportion of patients with clinically relevant shift (≤1.04 mmol/L)	13.4%	14.8%	13.3%	10.3%
Triglycerides, mmol/L				
Baseline	1.75 (1.06)	1.82 (1.07)	1.73 (1.16)	1.60 (0.74)
Change	0.20 (1.10)	0.25 (1.30)	0.16 (0.96)	0.11 (0.71)
Proportion of patients with clinically relevant shift (≥2.26 mmol/L)	15%	16.8%	13.5%	14.3%
Weight, Kg				
Baseline	82.5 (22.13)	80.0 (18.30)	84.1 (24.42)	86.8 (26.30)
Change	0.4 (5.16)	0.5 (5.41)	0.3 (5.32)	0.2 (3.97)
Patients with ≥7% increase in body weight, (%)				
Proportion of patients (%)	12.9	11.7	14.9	12.2
Waist/hip ratio				
Baseline	0.98 (0.22)	0.97 (0.19)	1.00 (0.29)	0.95 (0.09)
Change	0.01 (0.07)	0.01 (0.07)	0.01 (0.06)	−0.00 (0.08)

[a]Values shown as mean (SD) unless otherwise indicated and based on non-missing data.
[b]The *n* values are for the complete safety analysis set; number of patients varied for each parameter based on non-missing data.
[c]One patient with no reported reason for switching to quetiapine XR is included in the Overall column.
[d]This status was determined based upon a documented report from the patient that last meal was >8 hours before blood sample taken for baseline and post-baseline laboratory measurements. However, not all samples could be confirmed as fasted.

highest rate of any subgroup, but the sample size was small ($n=16$).

The findings for the primary endpoint were supported by the results of numerous secondary endpoints with change in CGI-CB, CGI-I, and CGI-S scores being significantly improved for the overall study population (all $p<0.0001$). Similarly, the significant improvements on PANSS total score, and positive, negative, and general psychopathology symptoms subscale scores (all $p<0.001$), as well as the improvements in GAS and SOFAS scores (both $p<0.0001$), all supported the primary efficacy outcome for the overall study population. Defined symptomatic remission was also achieved and/or maintained in a majority of patients at the 6 month visit (detailed in separate publication). Taken together, these results suggest that quetiapine XR is an effective therapeutic option for the treatment of schizophrenia, and may also potentially enhance patients' social and occupational functioning.

Antipsychotic switch studies tend to be positive for the agent being switched to given the reasons that lead to the decision to switch in the first place. Clearly, the open-label design and lack of a placebo group in the present study do constitute further potential limitations; however, this was a multicentre, large scale, international study. Further, the naturalistic design allows clinically relevant conclusions to be drawn when treating outpatients with schizophrenia. In addition, the efficacy of quetiapine XR in patients with schizophrenia has already been established in previous randomized, double-blind, placebo-controlled, registration trials[12–14], and in open-label trials of shorter duration[18]. Another limitation of the present study relates to the small sample size of some of the subgroups that may limit conclusions drawn from subgroup analyses. However, the main goal of the present study was to demonstrate the clinical benefit of switching to quetiapine XR in the overall population and the study was not powered to examine the effects in different subgroups. The analyses based on

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reason for switching or prior antipsychotic medication provide additional information to guide clinicians as to which patients are most likely to benefit from switching to quetiapine XR and promote further research on the efficacy and safety of quetiapine XR in specific subgroups of patients.

The titration of quetiapine XR to therapeutic dose within 3 days was derived from the product monograph and followed the titration used in other studies with quetiapine XR[14,18,26]. Extended release, compared to immediate release, formulations allow for more rapid titration primarily due to improved tolerability[27]. The ability to actually achieve therapeutic dose (and rapidly) potentially leads to greater initial response which has been shown to correlate with later efficacy and improved outcome[27–31]. There was a rapid decrease and discontinuation of prior antipsychotics as in the short-term study[18], but this study also confirmed that a short cross-taper could be successfully achieved and avoided exposing patients to a lengthy combination of antipsychotics that can lead to failed switches and more adverse effects[32]. Furthermore, depending on the antipsychotic being switched, little difference has been shown between different switching strategies[33,34]. In this study, 27.5% of patients showed improvement in CGI-CB scores from week one (decrease of at least one point) suggestive of early efficacy and tolerability.

The discontinuation rate of 38% likely reflects the fact that this was a long-term (for schizophrenia studies), monotherapy study in which patients with at least moderately severe schizophrenia (baseline mean PANSS total score of 74.5 [22.5]; baseline mean CGI-S of 3.8 [1.2])[35], were all treated with one agent according to a fixed titration and dose schedule. Patients prior to entry could have been previously treated with any antipsychotic including clozapine, LAIs, combination antipsychotics, etc. In a naturalistic study with a real world population of patients with schizophrenia it would be expected that a proportion of patients will discontinue for many different reasons. Further, not all patients are going to respond favourably to a trial of one particular antipsychotic, in this case quetiapine XR, and/or in the dose range allowed.

Patients previously on quetiapine IR demonstrated a significant improvement in CGI-CB on comparable dosing, 544 mg (219.6) for quetiapine IR vs. 558 mg (311.2) for quetiapine XR. It is postulated that with the once-daily XR formulation there are not only more stable plasma concentrations but also a differential occupancy of dopamine D2 receptors over a 24 hour period compared with quetiapine IR given twice daily[36,37]. There are no differences in exposure as the AUC and C_{max} are the same[26,38]. Patients in this study also demonstrated a high level of adherence which may be due at least in part to the simple titration regimen and/or the once-daily administration schedule as well as participation in a study. In addition, medication was supplied in blister packages and reviewed at every study visit. The tolerability of quetiapine XR is reflected in the low number of AEs observed during the 24 week study and also consistent with the known side effect profile of quetiapine XR[12–14].

Side effect profile is an important consideration when choosing an antipsychotic agent particularly with respect to movement disorders and metabolic disturbances. The majority of patients who switched to quetiapine XR in this study showed stable or improved scores on the BARS and SAS, indicating that quetiapine XR did not induce or exacerbate EPS. This lack of movement disorder has been previously reported for quetiapine IR compared to other SGAs, e.g. risperidone[39,40], and FGAs, e.g. haloperidol[41,42]. Although 12.9% of patients demonstrated an increase in weight of ≥7%, mean overall weight gain was 0.4 kg which compares with 1.04 kg observed in a 6 month relapse prevention study with quetiapine XR[15]. In addition, switching to quetiapine XR resulted in a slightly decreased weight in the olanzapine subgroup, which is consistent with reports from other studies evaluating switching to quetiapine IR[43,44]. It could be argued that patients previously treated with antipsychotics may have already reached a plateau in terms of weight gain or that 6 months may have been insufficient to show change. However, metabolic changes are observed within 2 weeks of switching to certain SGAs such as olanzapine with 55.4% of adult patients showing ≥7% weight gain over 24 weeks[45,46]. Recent data from the Schizophrenia Outpatient Health Outcomes (SOHO) study showed continued weight gain over 3 years for all antipsychotics but the proportion of patients losing ≥7% baseline weight was greatest for quetiapine[47]. Overall in the present study, there were minimal changes in other metabolic measures including WHR, BMI, FPG, fasting insulin, HbA1c, and cholesterol levels for the majority of patients. A total of 15% of patients showed a clinically relevant shift of triglycerides from baseline suggesting that some patients are more sensitive to metabolic effects which have been shown to be related to multiple factors including BMI at baseline, pharmacogenetic factors and previous treatment history[48]. In this study: mean triglyceride levels were already above normal at baseline; more than two thirds of patients were either overweight, obese or severely obese by BMI at baseline (mean BMI = 28.7 [6.89] kg/m^2); and more than 40% of patients were switching because of weight gain on prior antipsychotic (insufficient tolerability subgroup). Thus, in general, switching to quetiapine XR did not further worsen cardiometabolic risk factors in this study population that is at increased risk[49–52].

Conclusions

The domains of efficacy, safety and tolerability, and functionality contribute to the concept of clinical effectiveness[53,54]. The results of this study show that a majority of

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patients switched from other antipsychotics, including other SGAs, to quetiapine XR experienced improved effectiveness as measured by a composite outcome of clinical benefit (CGI-CB scale) that comprises both efficacy and burden due to side effects. Improvement in efficacy was supported by all efficacy measures (CGI-I, CGI-S, PANSS), regardless of the reason for switching. In addition, the type, pattern, and severity of AEs, as well as laboratory measures including cardiometabolic, demonstrated that quetiapine XR was generally safe and well tolerated in patients with schizophrenia managed in a clinical setting.

Transparency

Declaration of funding

This study (D1443L00025) was funded by AstraZeneca Canada Inc. P.C., A.G. and C.S. conceived and designed the study. P.C. was the international principal investigator (PI). With the exception of P.C. and the Astra Zeneca employees (A.G., C.S., H.E.) all the other authors were site PIs in their respective countries. P.C., A.G. and C.S. drafted the initial manuscript with editorial assistance from Christina Clark. P.C., A.G. and C.S. interpreted the results with statistical assistance from Robert Balshaw. All the authors contributed to interpretation of the results and drafting of the manuscript. All the authors reviewed and approved the final version of the manuscript.

Declaration of financial/other relationships

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